4. **Scenario: You are working on a project that involves analyzing the sales performance of a**

company over the past four quarters. The quarterly sales data is stored in a NumPy array named

sales_data, where each element represents the sales amount for a specific quarter. Your task is to

calculate the total sales for the year and determine the percentage increase in sales from the first

quarter to the fourth quarter.

Question: Using NumPy arrays and arithmetic operations calculate the total sales for the year and

determine the percentage increase in sales from the first quarter to the fourth quarter?

Code;

```
import numpy as np


# Quarterly sales (Q1, Q2, Q3, Q4)

sales_data = np.array([250000, 280000, 310000, 360000])


# Total sales for the year

total_sales = np.sum(sales_data)


# Percentage increase from Q1 to Q4

q1 = sales_data[0]

q4 = sales_data[3]

percentage_increase = ((q4 - q1) / q1) * 100


print("Quarterly sales:", sales_data)

print("Total sales for the year:", total_sales)

print("Percentage increase from Q1 to Q4: {:.2f}%".format(percentage_increase))
```

output;

Quarterly sales: [250000 280000 310000 360000]

Total sales for the year: 1200000

Percentage increase from Q1 to Q4: 44.00%

5. Scenario: You are a data analyst working for a car manufacturing company. As part of your

analysis, you have a dataset containing information about the fuel efficiency of different car

models. The dataset is stored in a NumPy array named fuel_efficiency, where each element

represents the fuel efficiency (in miles per gallon) of a specific car model. Your task is to calculate

the average fuel efficiency and determine the percentage improvement in fuel efficiency between

two car models.

Question: How would you use NumPy arrays and arithmetic operations to calculate the average

fuel efficiency and determine the percentage improvement in fuel efficiency between two car

models?

Code;

```
import numpy as np


# Fuel efficiency (miles per gallon) of 6 car models
fuel_efficiency = np.array([28.5, 32.0, 35.5, 30.0, 40.0, 26.0])


# Average fuel efficiency
average_mpg = np.mean(fuel_efficiency)
```

```
# Compare two car models: Model 1 (index 0) and Model 5 (index 4)
old_model = fuel_efficiency[0]
new_model = fuel_efficiency[4]
improvement = ((new_model - old_model) / old_model) * 100

print("Fuel efficiency (mpg):", fuel_efficiency)
print("Average fuel efficiency: {:.2f} mpg".format(average_mpg))
print("Model 1 = {} mpg, Model 5 = {} mpg".format(old_model, new_model))
print("Percentage improvement from Model 1 to Model 5:
{:.2f}%".format(improvement))
```

output;;

Fuel efficiency (mpg): [28.5 32. 35.5 30. 40. 26.]

Average fuel efficiency: 32.00 mpg

Model 1 = 28.5 mpg, Model 5 = 40.0 mpg

Percentage improvement from Model 1 to Model 5: 40.35%

6. Scenario: You are a cashier at a grocery store and need to calculate the total cost of a customer's

purchase, including applicable discounts and taxes. You have the item prices and quantities in

separate lists, and the discount and tax rates are given as percentages. Your task is to calculate the

total cost for the customer.

Question: Use arithmetic operations to calculate the total cost of a customer's purchase, including

discounts and taxes, given the item prices, quantities, discount rate, and tax rate?

Code;

Item prices and quantities in separate lists

```
prices = [50, 120, 30, 75]
quantities = [2, 1, 5, 3]

discount_rate = 10   # percent
tax_rate = 8       # percent

# Subtotal = sum of price * quantity
subtotal = 0
for price, qty in zip(prices, quantities):
   subtotal += price * qty

# Apply discount
discount_amount = subtotal * discount_rate / 100
after_discount = subtotal - discount_amount

# Apply tax on the discounted amount
tax_amount = after_discount * tax_rate / 100
total_cost = after_discount + tax_amount

print("Subtotal         :", subtotal)
print("Discount ({}%)     : {:.2f}".format(discount_rate, discount_amount))
print("After discount     : {:.2f}".format(after_discount))
print("Tax ({}%)        : {:.2f}".format(tax_rate, tax_amount))
print("Total cost to pay   : {:.2f}".format(total_cost))
output;
Subtotal         : 595
Discount (10%)     : 59.50
After discount     : 535.50
```

Tax (8%) **: 42.84**

Total cost to pay **: 578.34**
